

AM5920 - Deep learning for Mechanics

Assignment No. 2, Due Date: 5th April, 2026

1. Consider the following equation of wave propagation in rod:

$$\frac{\partial^2 u}{\partial t^2} = \frac{\partial}{\partial x} \left(E(\epsilon) \frac{\partial u}{\partial x} \right)$$

with initial and boundary conditions

$$\begin{aligned} u(x, 0) &= u_0(x), \quad \frac{\partial u}{\partial t}(x, 0) = v_0(x), \\ u(0, t) &= 0, \quad u(L, t) = 0. \end{aligned}$$

where $u(x, t)$ is the displacement field, $\epsilon(x, t) = \frac{\partial u}{\partial x}$ is the strain field and E is the Young's modulus that is a nonlinear function of strain ϵ . Let us assume that we measured displacement field $u(x_j, t_j) = \tilde{u}_j$ at N_x data points in the domain $[0, L]$ and N_t time snapshots. Now let us try to infer the Young Modulus at those measurement points.

Choose your favorite functions for $u_0(x)$ and $v_0(x)$ and infer $E(\epsilon)$ following the method of Physics informed neural network.

2. Consider the following advection-diffusion equation

$$\frac{\partial u}{\partial t} + a \frac{\partial u}{\partial x} - \epsilon \frac{\partial^2 u}{\partial x^2} = 0, \quad x \in \Omega, \quad t > 0 \quad (1)$$

$$u(x, 0) = f(x), \quad x \in \Omega \quad (2)$$

$$u(x, t) = g(x), \quad x \in \partial\Omega, \quad t > 0 \quad (3)$$

(i) Advection-dominated regime: $\epsilon = 0$ and $a = 0.1, 1, 10$. Investigate the gradient pathology by computing the eigenvalues of the Kernel terms corresponding to the boundary and PDE loss terms. How to handle spectral bias in this case?

(ii) Diffusion-dominated regime: $\epsilon = 10$ and $a = 0.1$. Investigate the gradient pathology by computing the eigenvalues of the Kernel terms corresponding to the boundary and PDE loss terms. Is spectral bias relevant for this case?

(iii) General: $\epsilon = 1$ and $a = 1$. Investigate the gradient pathology and spectral bias and mitigate them.